

Demonstrating NeuroFlux: A Non-Invasive Peripheral Magnetic Stimulation Device for Multimodal Haptic Feedback

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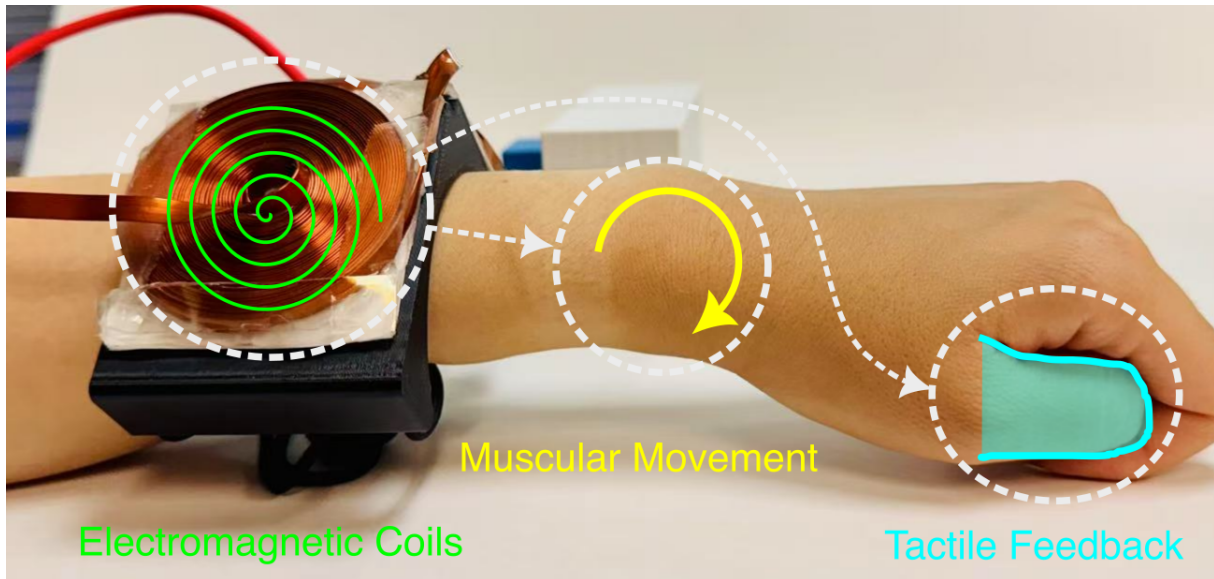


Figure 1: NeuroFlux is a wearable armband device using peripheral magnetic stimulation to generate multimodal haptic feedback, including muscular movement and tactile sensations.

Abstract

We demonstrate *NeuroFlux*, a wearable armband that delivers multimodal haptic feedback through non-invasive peripheral magnetic stimulation. Unlike conventional haptic devices limited to either tactile or kinesthetic modalities, *NeuroFlux* stimulates peripheral nerves to independently evoke both muscle movements and localized skin sensations. Our system features a custom-designed control circuit and a multi-coil armband, enabling precise, real-time

control of stimulation location and intensity. This hardware innovation significantly expands the design space of haptic feedback by bridging kinesthetic and tactile modalities through a single, compact device. In our demonstration, participants will experience a wide range of magnetically induced haptic sensations, including independent stimulation of muscular and cutaneous nerves in the forearm. The setup includes interactive tasks that showcase *NeuroFlux*'s ability to generate diverse haptic effects such as finger flexion, wrist movement, as well as immersive virtual reality object interactions. By offering hands-on exposure to peripheral magnetic stimulation, we aim to spark new research directions in multimodal haptic feedback and make neural stimulation more accessible to the HCI community.

CCS Concepts

• Human-centered computing → Haptic devices.

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1 Introduction

Haptic feedback has been a significant part of interactive experiences in extended reality [5], assistive technologies [8], immersive art [2], and robot teleoperation [7], etc. Existing haptic feedback devices all fall into two categories: tactile feedback and kinesthetic feedback [1]. Tactile feedback devices utilize mechanical indentations [18], vibrotactile actuators [6], peltier modules [10] and pneumatic / microfluidic chambers [14] to provide localized touch sensations on the skin. Kinesthetic feedback devices, on the other hand, involve electrical muscle stimulation [9], exoskeleton [11], and grounded force feedback controllers [16].

While these devices cover a wide range of haptic feedback modalities, each type of device is limited to only one modality. For example, vibrotactile devices are designated to produce high-frequency vibrations; Pneumatic devices are designated to generate skin indentations; And Exoskeletons are designated to limit limb movements and generate force feedback. While electrical muscle stimulation is believed to trigger both force and tactile feedback, the tactile tingling sensation is more like a "side effect" of the force feedback and cannot be controlled independently. In previous attempts to integrating multiple types of actuators into one platform [15], researchers discovered that a significant amount of effort is required even the integration only involved four tactile modalities. Till this date, there lacks an effective solution to providing kinesthetic and tactile feedback using one set of hardware configurations.

In this demonstration, we present *NeuroFlux*, a wearable haptic feedback armband that uses non-invasive peripheral magnetic stimulation to trigger both kinesthetic movements and tactile sensations. *NeuroFlux* was designed following the idea that all types of haptic feedback are transmitted to and from the central nervous system via peripheral nerves. Thus, kinesthetic feedback and tactile feedback can be triggered independently through precise control of stimulation points on different neural pathways.

Compared with previous HCI work that uses existing medical-grade magnetic stimulation devices [17], we designed and built our own peripheral magnetic stimulation device, with reduced form factor, improved wearing comfort, customized coil design, and precise control over stimulation parameters. *NeuroFlux* consists of two parts, the desktop control circuit, and the multi-coil armband. The control circuit receives Bluetooth trigger commands from the interaction program, and activates the corresponding electromagnetic coil. The coil generates a transient magnetic field and induces eddy currents inside the nerves underneath the skin. Depending on the type of nerves, the coil could either trigger muscle movements via muscular nerves, or trigger skin sensations via cutaneous nerves. Our evaluation demonstrates that *NeuroFlux* can independently

trigger muscular nerves and cutaneous nerves inside median, ulnar, and radial nerves of the forearm, producing a great variety of haptic feedback sensations on the hand.

The goal of this demonstration is to introduce the concept of peripheral magnetic stimulation to the UIST haptics community, and showcase the potential of neural stimulation to generate multimodal haptic feedback. We hope that by providing first-hand experience of magnetically induced haptic feedback sensations to researchers, this will inspire more ideas on novel applications of this emerging technology.

2 NeuroFlux Device Overview

We provide an overview of the *NeuroFlux* device implementation, including control circuit, electromagnetic coil, and mounting parts.

2.1 Control Circuit

Transcranial Magnetic Stimulation (TMS) works by discharging a high-current pulse through a wire coil to create a brief, rapidly changing magnetic field, which penetrates the scalp and skull without attenuation and induces a suprathreshold electrical current in the brain via electromagnetic induction [13]. This current can be delivered in monophasic or biphasic pulses, each affecting neuronal stimulation differently. Peripheral Magnetic Stimulation (PMS) operates on the same principle as TMS, using rapidly changing magnetic fields generated by high-current pulses through a coil to induce electrical currents in peripheral nerves or muscles. These induced currents can activate neuromuscular pathways noninvasively, enabling stimulation of limbs or targeted body regions without direct skin contact.

Peripheral magnetic stimulation requires the control circuit to discharge a significant amount of current (1kA) in a very short pulse (400us). To achieve this, the main components of the control circuit include a high-voltage DC-DC power converter and a charge-discharge control circuit (Figure 2). the power converter takes 15V power bench supply as input, and outputs 750V to the charge circuit. The control circuit uses the high voltage input to slowly charge up two 150uF capacitor, and when it receives a triggering command, it turns on the IGBT, and activates the coil.

2.2 Electromagnetic Coil Design

Electromagnetic coils are designed to focus on individual nerves in the arm, in order to trigger different kinesthetic feedback and tactile sensations on the hand. This requires customized coils to adapt to different nerve positions and depths. In prior work, researchers used electromagnetic simulations to evaluate the focality and depth of various coil designs [4]. Simulation results showed that Figure-8 coils, with two spiral coils connected in a "8" shape, have the best performance in both metrics. However, these prior evaluations are targeting transcranial magnetic stimulation (TMS) and designs are based on the human skull model. For our peripheral stimulation device, nerves are much more superficial underneath the skin and the major nerve bundles are more spatially separated from each other. Therefore, coils can be designed with smaller diameters, less turns, and more curvature.

Our coil is a Figure-8 shape coil with two spiral coils connected together (Figure 3). Each spiral coil uses 40 turns of rectangular

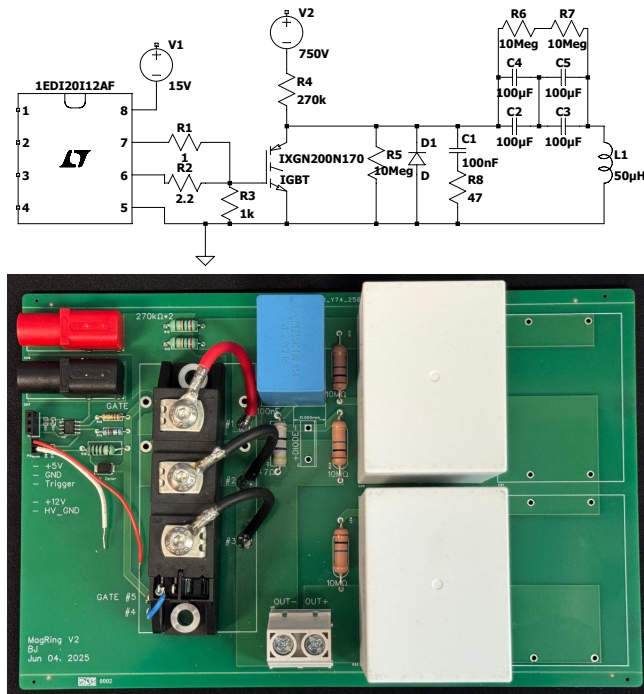


Figure 2: Top: Schematic diagram of the control circuit design. Bottom: Top-down view of the actual control circuit.

copper wires, with inner diameter of 10 mm and outer diameter of 60 mm. The coils were wound in opposite direction so that the magnetic fields complement each other. Unlike classic TMS coils that use a between-coil angle of 120-150 degree, the two spiral coils in our coil have an angle of 90 degree, wrapping more tightly around the arm surface and forming a strong magnetic gradient in the deeper tissue. This allows us to stimulate muscular nerves, which are innervated deeper in the muscle tissues. Electromagnetic field simulation in Ansys Maxwell (Figure 3) showed that with 1kA current input, the coil can generate up to 1.5T magnetic flux in the center, surpassing nerve activation thresholds [13].

2.3 Mounting Parts

We also designed a customized armband to mount the coil onto users' arms (Figure 4). The armband has a round curvature inside to fit users' arms and two flat surfaces outside to attach the Figure-8 coils. Inside the structure, we created two thin tunnels to allow stretchable round strings to go through, and the extruded strings are connected to a cord lock and can be adjusted as needed. This mounting mechanism ensures that the armband fits perfectly to arms with different curvatures and sizes, providing a consistent experience across users.

3 Demonstration Design

An interactive environment (Figure 5) will be setup to allow audience to experience NeuroFlux. The goal of this demonstration is to introduce the concept of peripheral magnetic stimulation to

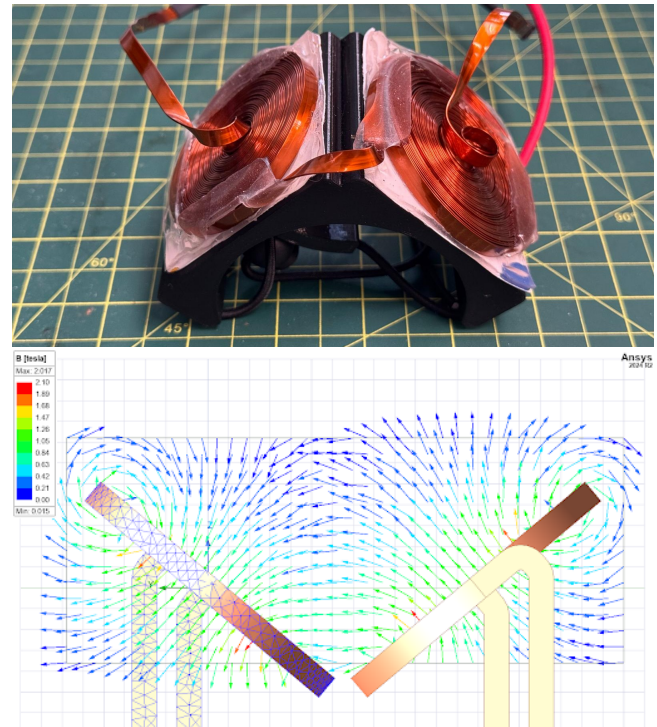


Figure 3: Top: Front view of the Figure-8 coil design. Bottom: Magnetic flux visualization of the Ansys electromagnetic field simulation.



Figure 4: NeuroFlux does not require skin contact and can be mounted on top of garments. The tightness of the armband can be adjusted with cord locks to adapt to different users.

the UIST haptics community, and showcase the potential of neural stimulation to generate multimodal haptic feedback.

3.1 Demonstration Space Layout

The demonstration area will be organized with several key components to facilitate interaction and learning:

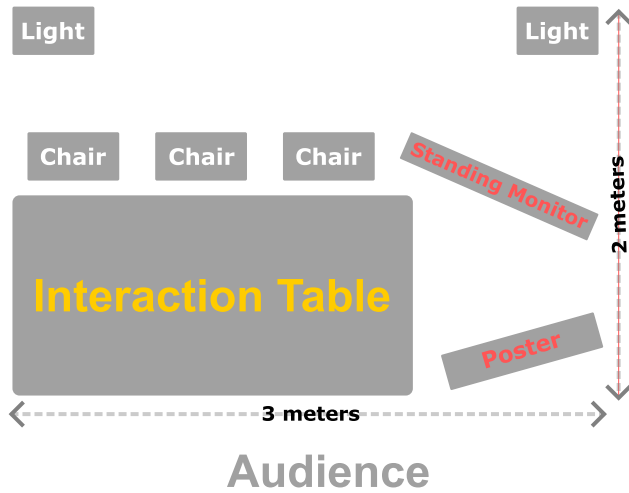


Figure 5: Floor plan of the demonstration setup. Interaction mainly happens at the table, while the poster and the monitor serve the purpose of attracting audience.

- Project poster: A large, visually appealing poster will be prominently displayed to provide a quick overview of the project, highlighting its key features and benefits. This poster will serve as an initial engagement point for attendees.
- Introduction video: A monitor will play a looping video that introduces the neural mechanisms behind the peripheral magnetic stimulation, implementation of the NeuroFlux device, and recordings showing NeuroFlux activating different nerves and inducing muscular movements. The video will help in setting the context and enhancing understanding among the audience.
- Interaction area: The NeuroFlux device will be placed on a table, allowing audience to try it on. Currently the NeuroFlux device includes one power bench supply, one control circuit PCB board, and one mounting device with coils. The control circuit will be placed in an insulated acrylic box, to prevent participants from touching the high voltage components. Participants will only have access to the mounting coils, which are wrapped with electrically isolated epoxy resin to prevent accidental conduction.

3.2 Interactive Experience

To ensure safety and maximize engagement and demonstrate NeuroFlux’s capabilities, the interactive experience will contain the following steps:

- (1) Preparation: The participant will first sign a liability waiver, then the authors will help the participant to put on the device, and tighten the strap with cord lock.
- (2) Individual Calibration: The authors will begin the stimulation at relatively low voltage input (600V), and gradually increase the voltage until the participant reports that the tactile sensation is felt clearly. This voltage will be set as the detection threshold for the participants. The authors

will continue to increase the voltage to a level when the participant reports that it is too strong, which is set as the maximum threshold. Then the stimulation voltage during the experience is set as the average of the two thresholds.

- (3) Nerve Activation: The NeuroFlux device will be placed on the wrist near the median nerve during the calibration. This will trigger strong tactile sensations on the thumb, index, and middle fingers. Then the device will be moved to the ulnar nerve and radial nerve area of the wrist to trigger tactile sensations on other fingers. If the participant is comfortable with the tactile feedback, the device will be moved to lower arm near the elbow to trigger muscular movements, with wrist reflex, extension, and tilt movements.
- (4) Virtual Reality Experience: NeuroFlux will serve as the haptic interface to provide prompting haptic feedback for object interaction in VR. The participant wearing the VR headset will see several objects floating in front of the view. When the participant reaches their hands within the range of a red cube, the participant will feel that its hand is pushed back from the cube, indicating that the object is not touchable. When the participants grasps a green sphere, tactile sensations will be triggered in the touching area between the sphere and the hand, which means safe to touch.

3.3 Safety Measurements

Magnetic stimulation is a FDA approved technology widely used in clinical applications [3], reflecting its overall safety. While magnetic stimulation carries some minor risks such as skin tingling or potential interference with implanted medical devices. these effects are generally preventable through appropriate participant screening and careful control of stimulation settings. A safety and ethics report on TMS concluded that no persistent adverse effects—such as seizures—have been observed when the technique is used in individuals without epilepsy [12]. Prior HCI work using magnetic stimulation for haptic interactions [17] provides clear evidence that the technique is safe to operate on human subjects.

3.4 Expected Outcome

We expect that NeuroFlux will open up the world of neural stimulation to HCI and haptics researchers. We hope that by providing first-hand experience of magnetically induced haptic feedback sensations to researchers, this will inspire more ideas of novel applications in haptics, sensory augmentation, healthcare and human-robot interaction, etc.

4 Conclusion

This demonstration presents a novel haptic interaction device, NeuroFlux, which leverages peripheral magnetic stimulation to render multimodal haptic feedback on the hand. Through an interactive demonstration space, participants will gain firsthand experience of the magnetically induced force and tactile feedback. This demonstration aims to enhance the communication between HCI and haptic researchers and spark potential collaborations in future projects.

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